

```

root@docker-master:/home/osboxes# docker run nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
54fec2fa59d0: Pull complete
4ede6f09aefe: Pull complete
f9dc69acb465: Pull complete
Digest: sha256:404ed8de56dd47adadad9e2641b1ba6ad5c
Status: Downloaded newer image for nginx:latest

```

Figure 3: Creating NGINX as a sample Docker container application

When we run the above command, Docker looks for the image repository and downloads the latest version of NGINX. Thus, the NGINX container is created and gets a virtual IP inside the private bridge network depicted in Figure 3.

The following command will display a list of all currently active containers. Successful deployment of NGINX Docker containers can be accessed as shown in Figure 4.

```
$ docker ps
```

Table 2: Frequently used commands for Docker in a production environment

Command	Purpose
\$ docker --version	To find the version of Docker
\$ "docker start "container_name"	Start your container
\$ "docker stop "container_name"	Stop running container
\$ docker kill "container_name"	Kill a container
\$ docker top "container_name"	To list the running processes within a container
\$ docker container stats "container_name"	To get the CPU and memory usage by a container
docker info	Get information on the container



Figure 4: NGINX Docker container successfully installed

Docker can provide a containerization platform that bundles our application and all of its dependencies inside a Docker container to ensure that our application functions seamlessly in any environment. Containerization simply eliminates this problem by bundling (containerizing) the software code with the related configuration files, libraries, and other run-time dependencies. Thus, containers are much more lightweight than virtual machines, resulting in their widespread adoption across industries. END 🐧

References

- <https://hub.docker.com/>
- <https://docs.docker.com/compose/compose-file/>

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